

A Perron-Theoretic Proof of the Polynomial Goldbach Theorem

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Abstract

In 1965, David Hayes showed that any polynomial $f \in \mathbb{Z}[X]$ of degree n can be expressed as the sum of two irreducible integer polynomials of degree n . In general it is true for any Noetherian ring R with infinitely many maximal R-ideals which was proved in the 21st century. Here I have used Perron's criterion of irreducibility to show the result for $R = \mathbb{Z}$.

1 Introduction

We all have definitely heard of the famous *Goldbach Conjecture*. It states that any even positive integer greater than 2 can be expressed as sum of two prime numbers. We say an element a in a ring R irreducible, if $a = bc$ implies either b or c is a unit (which has multiplicative inverse). In a Principal Ideal Domain (PID), every irreducible element is prime whose proof one can find in any algebra book. Since $\mathbb{Z}$ is a PID, every irreducible element is prime. Thus we can refine our statement of Goldbach conjecture by replacing prime with irreducible elements of $\mathbb{Z}$. Getting motivated from this problem, we ask the same question in polynomial ring. Can every polynomial over any arbitrary ring of degree n be expressed as the sum of irreducible polynomials of the same ring having degree n ? The answer is not always. But for Noetherian rings having infinitely many maximal ideals we can. Since $\mathbb{Z}$ is such a ring, for any $f \in \mathbb{Z}[X]$ there exists irreducible polynomials $g, h \in \mathbb{Z}[X]$ such that $f = g + h$ with $\deg(f) = \deg(g) = \deg(h)$. David Hayes proved it using Eisenstein's criterion of irreducibility. Here I will be proving this theorem using a very strong criterion for irreducibility called Perron's criterion which I will be proving.

2 Preliminaries

For this paper, one need to know some Complex analysis as the well known *Rouché's theorem* will be used in the proof of the Perron's criterion. I will not be proving this theorem as one can easily find it in any standard complex analysis book.

Theorem 2.1 (Rouché's theorem). *Let f and g be analytic functions on and inside a simple closed curve C . Suppose that $|f(z)| > |g(z)|$ for all points z on C . Then f and $f+g$ have the same number of zeros (counting multiplicities) interior to C .*

3 Perron's Irreducibility Criterion

Theorem 3.1 (Perron). *Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ be an integer polynomial with $a_0 \neq 0$ and*

$$|a_{n-1}| > 1 + |a_{n-2}| + \cdots + |a_1| + |a_0|$$

Then $P(x)$ is irreducible.

Proof. Let $\alpha \in \mathbb{C}$ be a root of $P(x)$ with $|\alpha| = 1$. Then

$$-a_{n-1}\alpha^{n-1} = 1 + \alpha^n + a_{n-2}\alpha^{n-2} + \cdots + a_0$$

$$|a_{n-1}| \leq 1 + |a_{n-2}| + \cdots + |a_1| + |a_0|$$

which contradicts the condition. Thus, $|\alpha| \neq 1$ or no roots lie on the unit circle. Now consider polynomials $f(x) = a_{n-1}x^{n-1}$ and $g(x) = P(x) - a_{n-1}x^{n-1}$ and the simple closed curve $\mathcal{C} = \{z \in \mathbb{C} : |z| = 1\}$. By the given condition we can easily say that

$$|f(z)| = |a_{n-1}z^{n-1}| > |P(z) - a_{n-1}z^{n-1}| = |g(z)|$$

for all $z \in \mathcal{C}$. Thus by Rouché's theorem, we can say that $a_{n-1}z^{n-1}$ and $P(z)$ has same number of zeroes inside $\mathcal{C}$. Since f has exactly $n-1$ zeroes inside, $P(z)$ also has exactly $n-1$ zeroes strictly inside $\mathcal{C}$ (as no roots can lie on $\mathcal{C}$) and another say α with $|\alpha| > 1$. Now suppose P is reducible, then there exists $f, g \in \mathbb{Z}[X]$ with $\deg(f), \deg(g) \geq 1$ such that

$$P(x) = f(x)g(x)$$

Since $P(0) = a_0 \neq 0$, $|f(0)|, |g(0)| \geq 1$. WLOG, let α be a root of $g(x)$. Let $\deg(f) = k < n$ and $\alpha_1, \alpha_2, \dots, \alpha_k$ be the roots of $f(x)$. Then $|\alpha_i| < 1$ for all $1 \leq i \leq k$. Also since $P(x)$ is a monic polynomial, f, g also has to be monic. Using Vieta's theorem we can say that

$$1 \leq |f(0)| = \left| \prod_{i=1}^k \alpha_i \right| < 1$$

which is a contradiction. Thus, P is irreducible. □

This criterion makes our life very easy while solving irreducibility problems. One can solve the following example problem directly by invoking the theorem.

Example 3.1 (IMO 1993). Let $f(x) = x^n + 5x^{n-1} + 3$, where $n > 1$ is an integer. Prove that $f(x)$ cannot be expressed as the product of two nonconstant polynomials with integer coefficients.

The following are some direct exercises to enjoy!

Exercise 3.1. Let $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$ be a polynomial with $a_i \in \mathbb{C}$ such that

$$2|a_k| > \sum_{i=0}^n |a_i|$$

for some $0 \leq k \leq n$. Then show that exactly k many roots of P are strictly inside the unit circle and the other $n - k$ outside the circle.

Exercise 3.2. Let $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$ be a polynomial with $a_i \in \mathbb{Z}$ and $|a_0|$ is prime such that

$$2|a_0| > \sum_{i=0}^n |a_i|$$

Show that P is irreducible in $\mathbb{Z}[X]$.

Exercise 3.3 (Iran 2003). Let $f_1, f_2, \dots, f_n$ be polynomials with integer coefficients. Show that there exists a reducible polynomial $g(x) \in \mathbb{Z}[x]$ such that $f_i(x) + g(x)$ is irreducible for $i = 1, 2, \dots, n$.

4 The Main Theorem

Theorem. For any polynomial $P \in \mathbb{Z}[X]$ of $\deg(P) \geq 1$, there exists $f, g \in \mathbb{Z}[X]$ irreducible polynomials with $\deg(P) = \deg(f) = \deg(g)$ such that

$$P(x) = f(x) + g(x)$$

Proof. In my proof I will be explicitly constructing polynomials f, g to get the theorem. Suppose $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ with $a_i \in \mathbb{Z}[X]$. Define

$$f(x) = b_n x^n + (a_{n-1} - pM)x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + (a_0 - p)$$

$$g(x) = c_n x^n + pMx^{n-1} + p$$

where p is some prime such that $p \nmid c_n$ and $a_0 - p \neq 0$.

If $a_n = 1$, $b_n = -1$ and $c_n = 2$ else $b_n = 1$ and $c_n = a_n - 1$. Now choose M such that $p \nmid M$ and

$$|a_{n-1} - pM| > 1 + |a_{n-2}| + \cdots + |a_1| + |a_0 - p|$$

Thus one can easily check that $P(x) = f(x) + g(x)$. Observe that g is an Eisenstein polynomial and f satisfies the Perron's criterion as per the construction, thus both f, g are irreducible. This completes the proof! $\square$

Example 4.1. Consider a polynomial $p(x) = x^5 + 4x^4 - 2x^2 + 7$. Using the above construction, we take the polynomials $g(x) = 2x^5 + 3 \cdot 8x^4 + 3$ and $f(x) = -x^5 - 20x^4 - 2x^2 + 4$ where both f, g are irreducible and $p = f + g$.

The one who are interested to explore more on this, can use the following references to get the proof of the general theorem.

References

- [1] D. R. Hayes, *A Goldbach Theorem for Polynomials with Integral Coefficients*, The American Mathematical Monthly, vol. 72, no. 1, 1965, pp. 45–46.
- [2] P. Pollack, *On Polynomial Rings with a Goldbach Property*, The American Mathematical Monthly, vol. 118, no. 1, 2011, pp. 71–77.